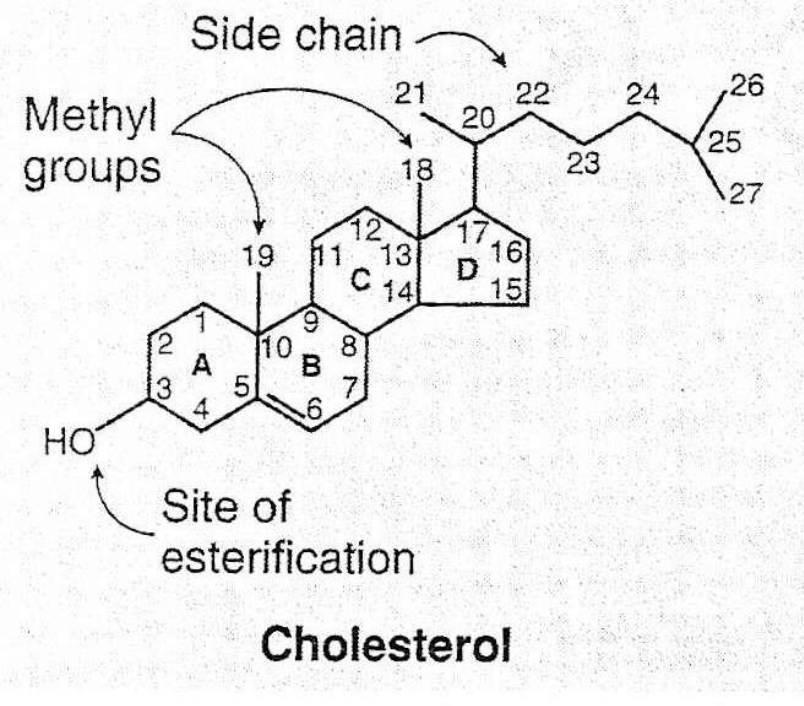
Cholesterol And Sterol Metabolism

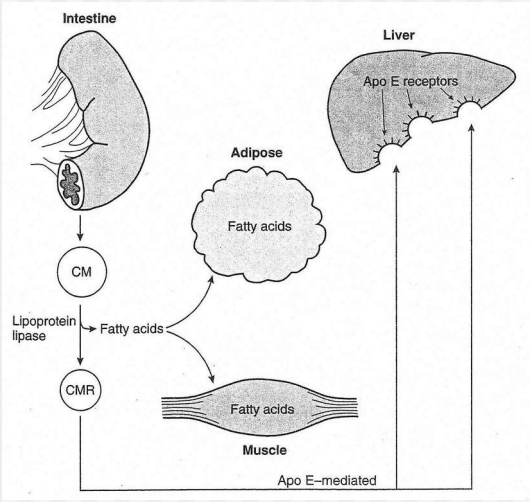
Cholesterol Functions

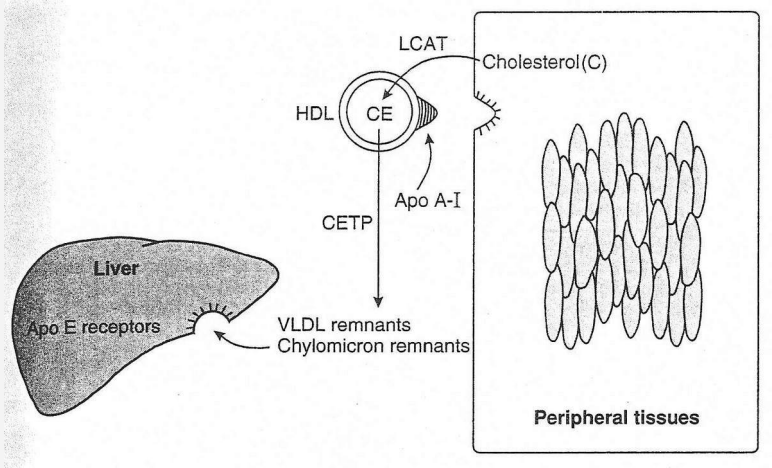
- Membrane component
- Precurser to
  - Bile acids
  - Vitamin D
  - Steroid hormones



Central Role of the Liver in Cholesterol Balance: Sources of hepatic cholesterol

- Dietary cholesterol
  - From chylomicron remnants
- Cholesterol from extra-hepatic tissues
  - Reverse cholesterol transport via HDL
    - Chylomicron remnants
    - IDL
- De novo synthesis



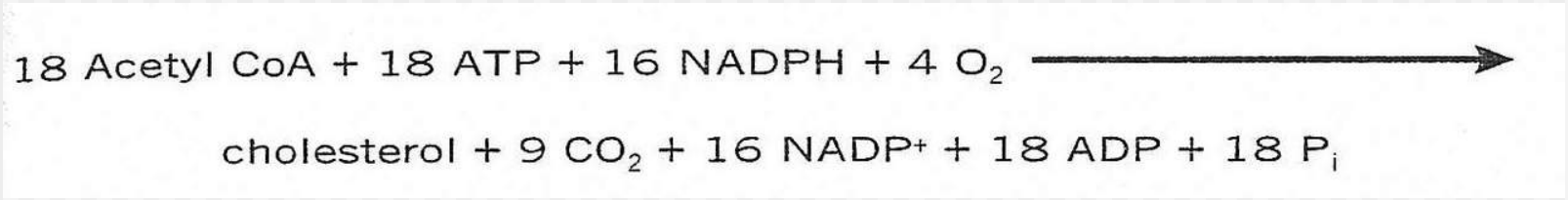


Central Role of the Liver in Cholesterol Balance:
Fate of hepatic cholesterol

- VLDL → LDL
  - Transport to extra-hepatic tissues
- Direct excretion into bile
  - Gallstones commonly are precipitates of cholesterol
    - Occurs when bile becomes supersaturated with cholesterol
      - Obesity, biliary stasis, infections
- Bile acid synthesis and excretion into bile

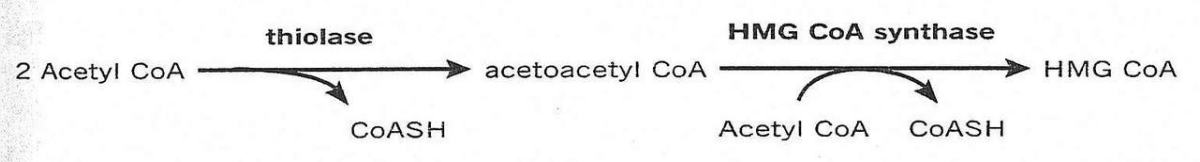
De novo Synthesis of Cholesterol

- Primary site: liver (~1g/d)
  - Secondary sites: adrenal cortex, ovaries, testes
- Overall equation:

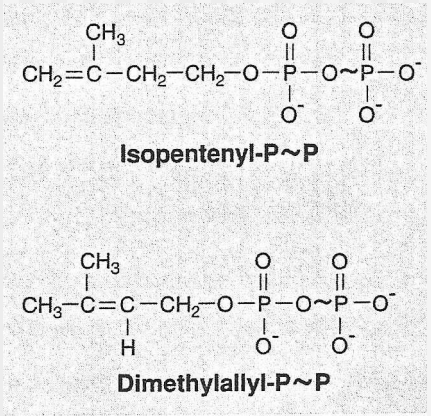


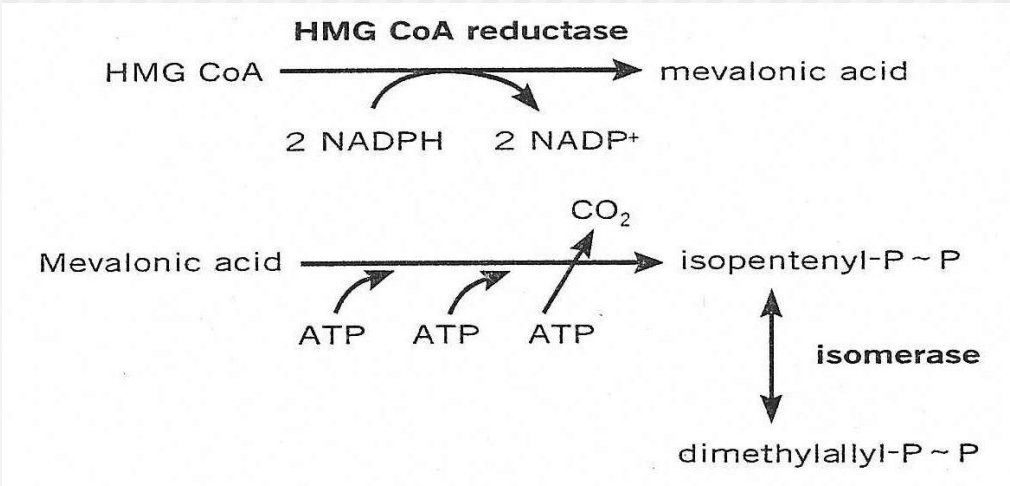
De novo Synthesis of Cholesterol:four stages

- Formation of HMG CoA (cyto)
  - Analogous to KB synthesis (mito)



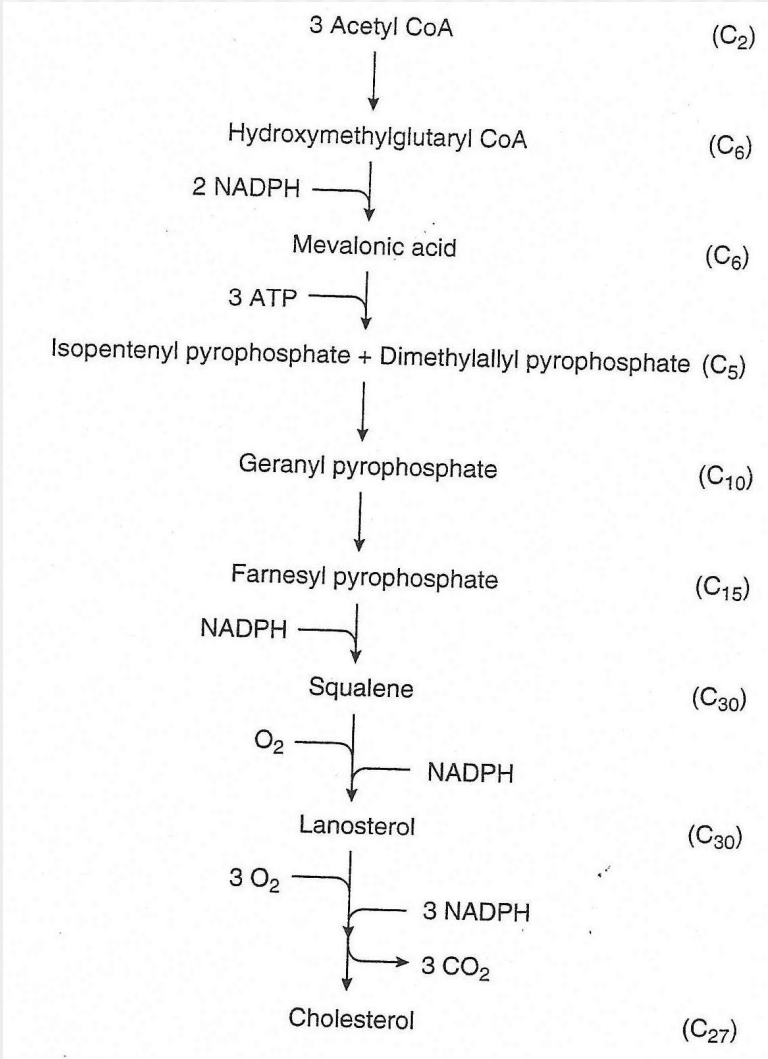
- Conversion of HMG CoA to activated isoprenoids





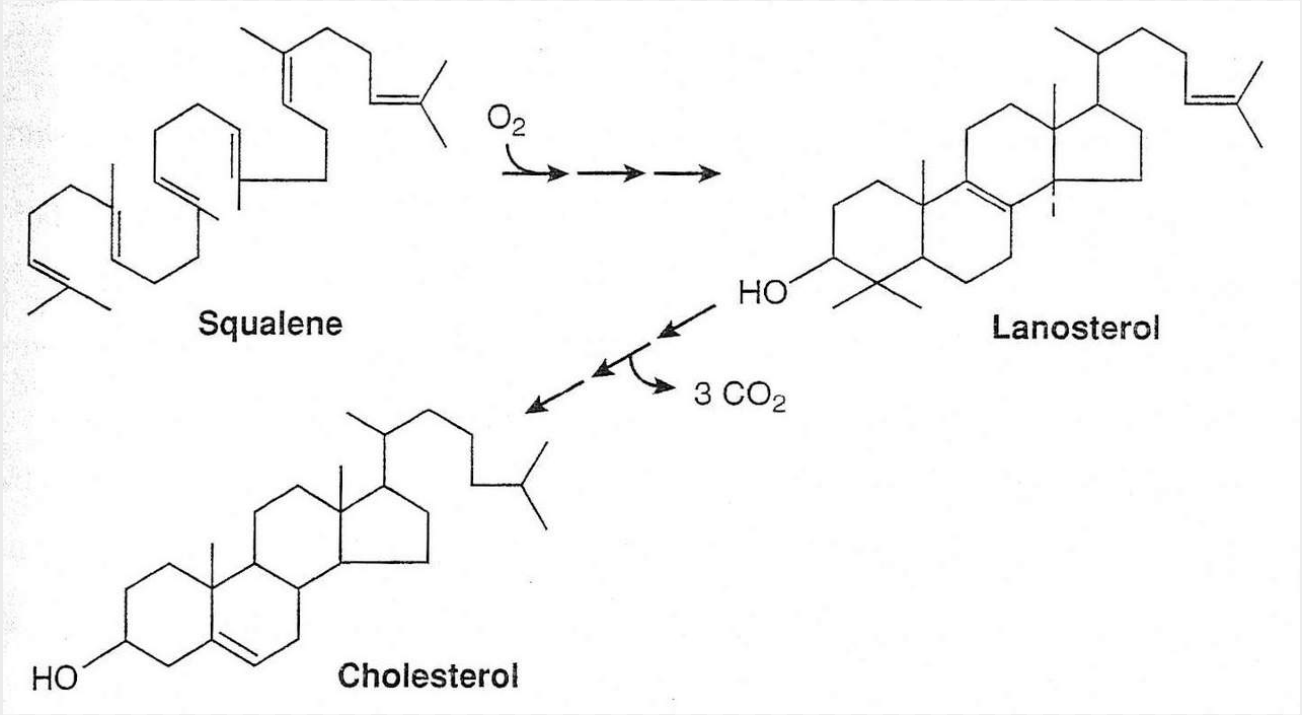
De novo Synthesis of Cholesterol:four stages

- Condensation of isoprenoids to squalene
  - Six isoprenoids condense to form 30-C molecue



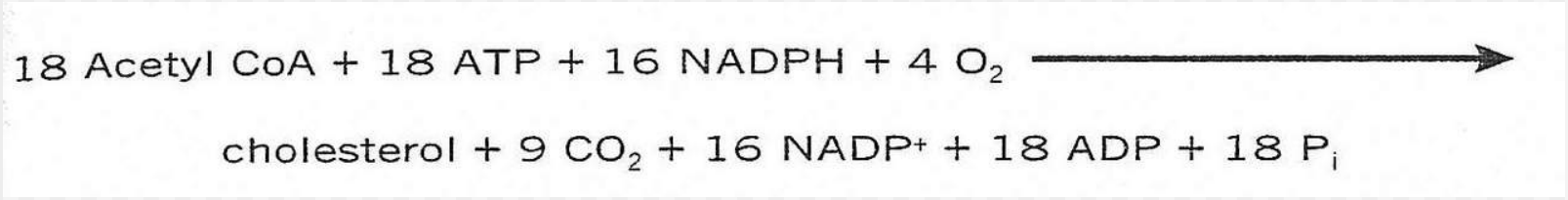
De novo Synthesis of Cholesterol:four stages

- Conversion of Squalene to Cholesterol



De novo Synthesis of Cholesterol: What do you need to know?

- All carbons from acetyl-CoA
- Requires NADPH, ATP, & O_2
- Stages
  - One: forms HMG CoA
  - Two: forms activated 5 carbon intermediates (isoprenoids)
  - Three: six isoprenoids form squalene
  - Four: squalene + O_2 form cholesterol



Smith-Lemli-Opitz (SLO) syndrome

- 3rd most common inborn error of metabolism in US
  - Cystic fibrosis & PKU
  - 1/30 Americans of N. European descent are carriers (heterozygous advantage??)
  - Prevalence of SLO is 1:20-60,000
    - Predicted prevalence: 1:5-18,000
    - Spontaneous abortion
    - Underdiagnosed ?
      - “multiple congenital abnormality syndrome of unknown aetiology”
- Deficiency of delta-7-dehydrocholesterol reductase
  - 7-dehydrocholesterol →desmosterol → cholesterol

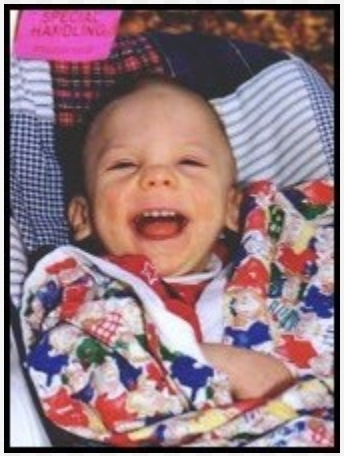
Smith-Lemli-Opitz (SLO) syndrome

- multiple congenital anomalies/mental retardation
  - Spontaneous abortions/still births
  - Multiorgan failure shortly after birth
  - Congenital heart disease: cyanosis or congestive heart failure
  - Vomiting, feeding problems, failure to thrive
  - Visual and hearing loss
- Pathophysiology
  - Very low plasma cholesterol
  - Inability to synthesize cholesterol
    - Membranes, precursers for steroid hormones & bile acids, myelin component

The following characteristics have been seen in more than 50% of patients:

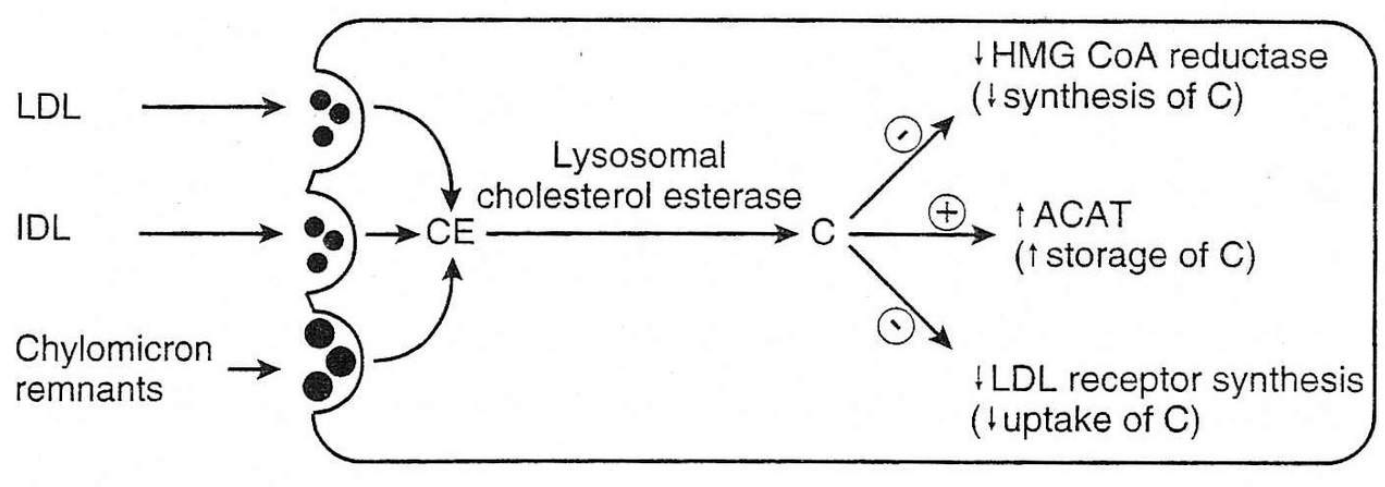
- Microcephaly
- Blepharoptosis (drooping of the upper eyelids)
- Cleft palate
- Postnatal growth retardation
- Syndactyly of toes (webbing between toes)
- Mental retardation
- Hypospadias (developmental anomaly involving the urethra)
- Hypotonia
- Inner epicanthal folds
- Low-set ears
- Small, upturned nose
- Small tongue
- Undescended testicles
- Micrognathia (small jaw)
- Broad maxillary alveolar ridges





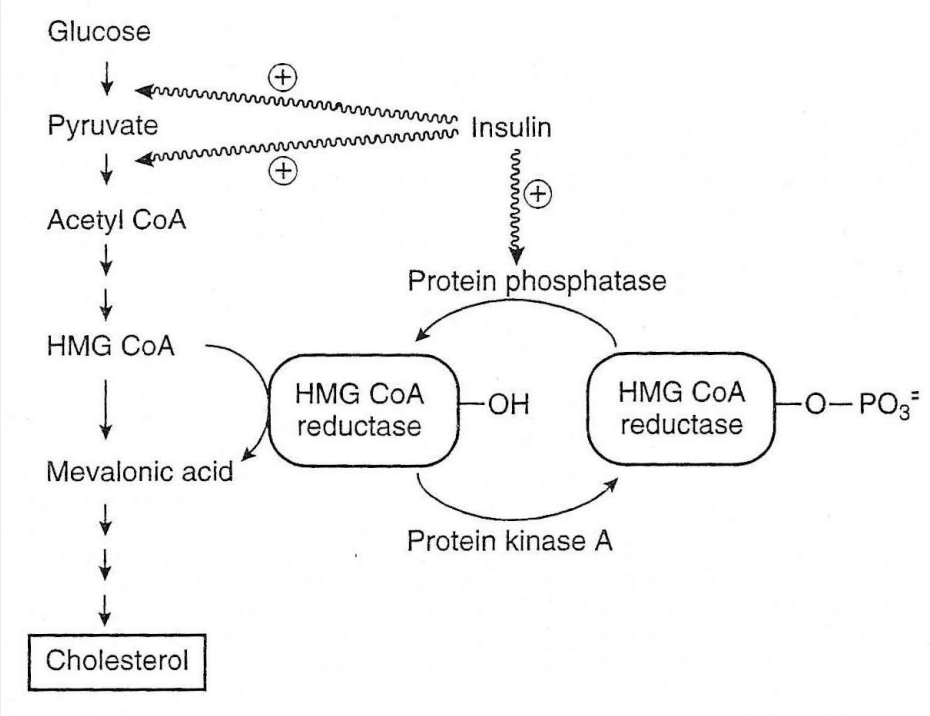
Regulation of Cholesterol Synthesis

- Cellular cholesterol content exerts transcriptional control
  - HMG-CoA reductase
    - Half life = 2 hours
  - LDL-receptor synthesis
- Nutrigenomics:
  - interactions between environment and individual genes and how these interactions affect clinical outcomes



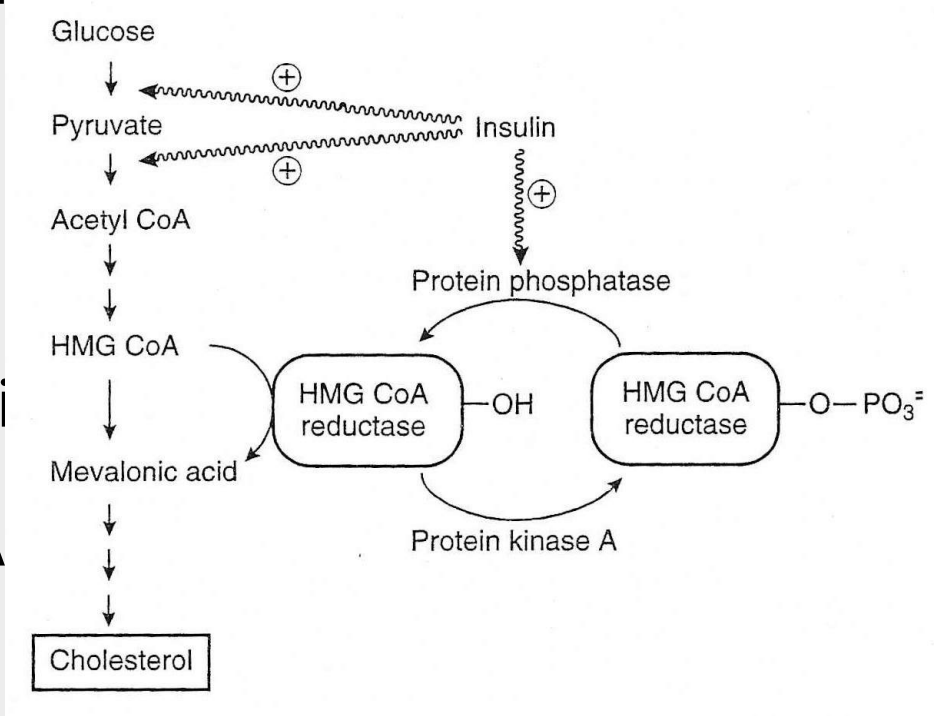
Regulation of Cholesterol Synthesis

- Covalent Modification of HMG-CoA Reductase
  - Insulin induces protein phosphatase
  - Activates HMG-CoA reductase
- Feeding promotes cholesterol synthesis
  - Activates reg. enzyme
  - Provides substrate: acetyl CoA
  - Provides NADPH



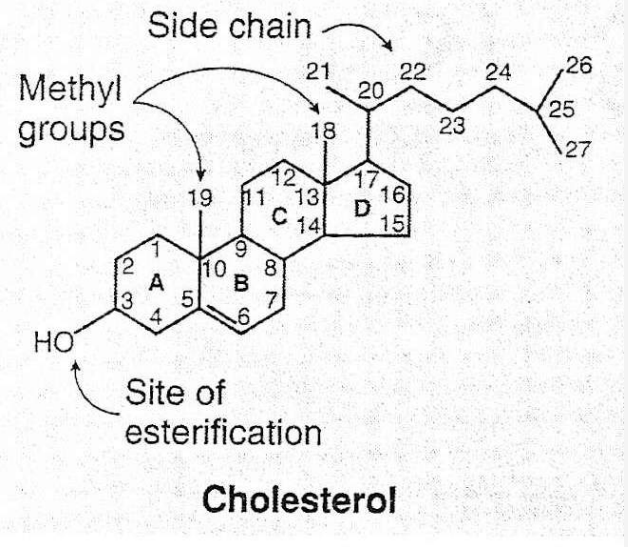
Regulation of Cholesterol Synthesis

- Covalent Modification of HMG-CoA Reductase
  - Glucagon stimulates adenyl cyclase producing cAMP
  - cAMP activates protei kinase A
  - Inactivates HMG-CoA reductase
- Fasting inhibits cholesterol synthesis

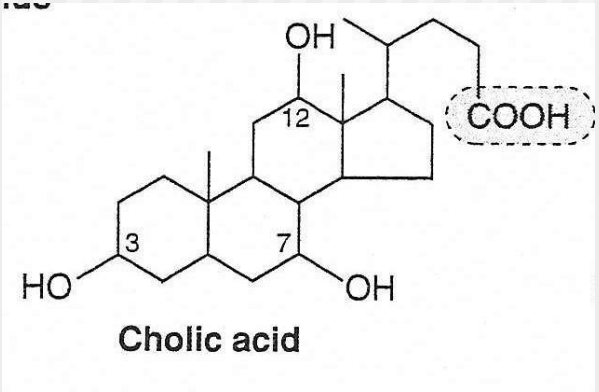


Cholesterol and Bile Acid/Salt Metabolism

- Major excretory form of cholesterol
  - Steroid ring is not degraded in humans
  - Occurs in liver
- Bile acid/salts involved in dietary lipid digestion as emulsifiers



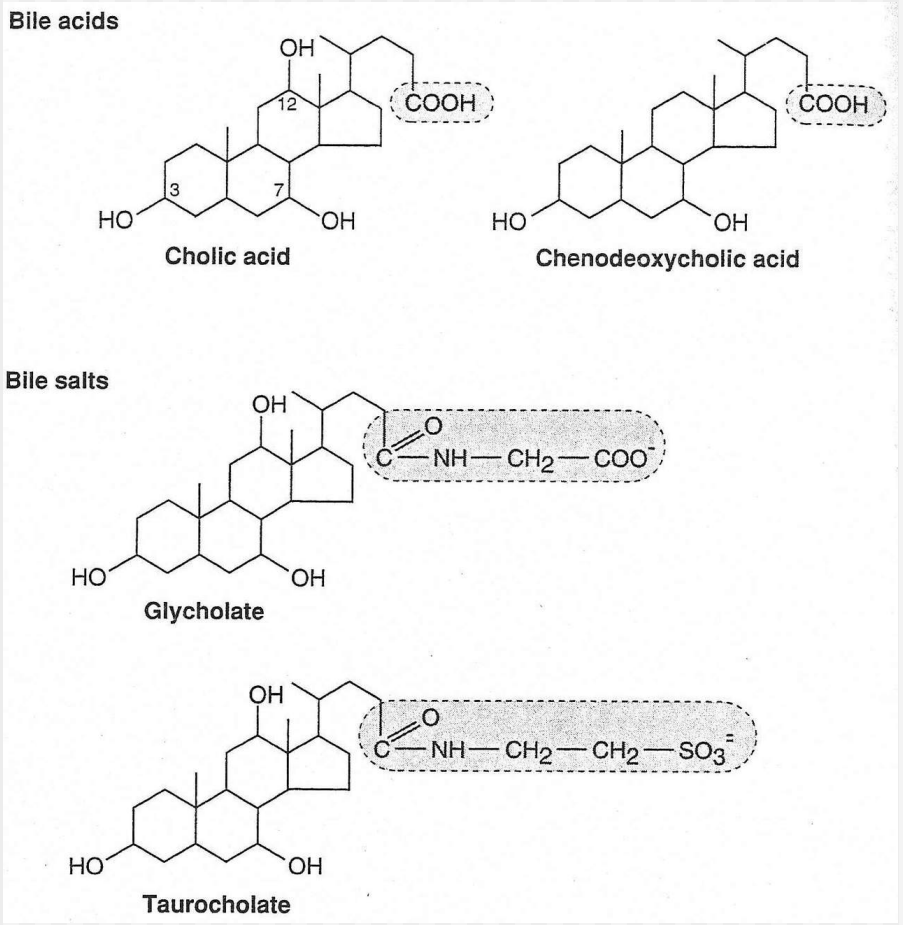
Cholesterol



Cholic acid

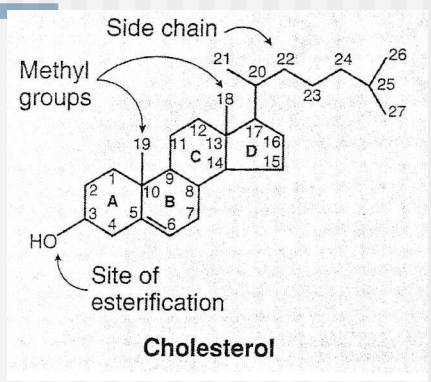
Types of Bile Acids/Salts

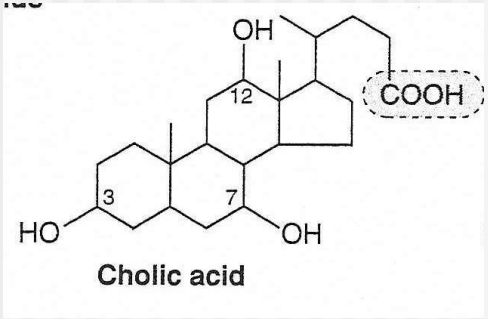
- Primary bile acids
  - Good emulsifying agents
    - All OH groups on same side
    - pKa = 6 (partially ionized)
- Conjugated bile salts
  - Amide bonds with glycine or taurine
  - Very good emulsifier
    - pKa lower than bile acids

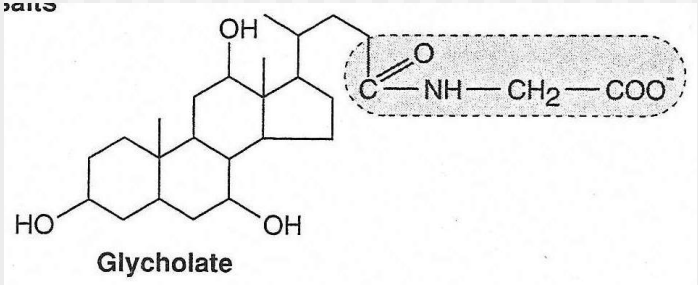


Synthesis of Bile Salts

- Hydroxylation
  - Cytochrome P-450/mixed function oxidase system
- Side chain cleavage
- Conjugation
- Secondary bile acids
  - Intestinal bacterial modification
    - Deconjugation
    - Dehydroxylation
      - Deoxycholic acid
      - Lithocholic acid

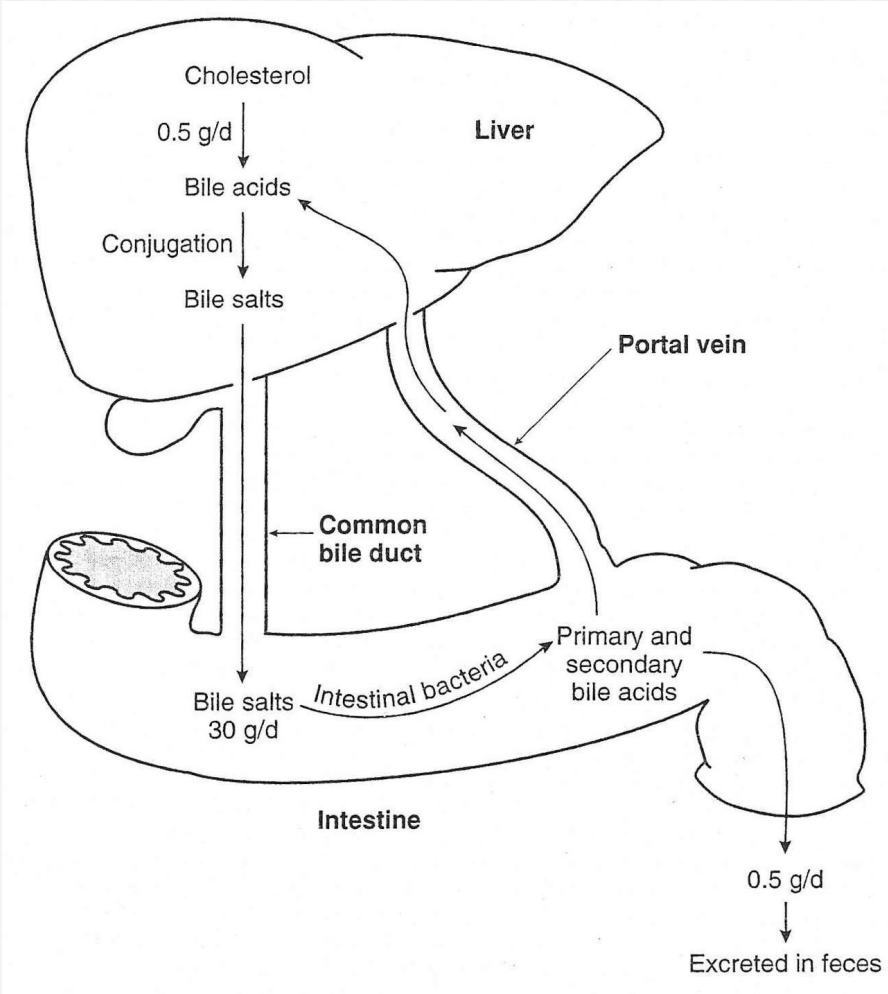






Recycling of Bile Acids

- Enterohepatic circulation
  - 98% recycling of bile acids
- Cholestyramine Treatment
  - Resin binds bile acids
  - Prevents recycling
  - Increased uptake of LDL-C for bile acid synthesis



Nutritional and Pharmaceutical Means for Treating Hypercholesterolemia

- Reducing intake of dietary saturated fat to < 7% of calories
  - Proposed mechanism:
    - High saturated fat intake reduces activity of LDL-receptors
    - Higher unsaturated fat intake increases activity of LDL-receptors
  - Side effects: none

Nutritional and Pharmaceutical Means for Treating Hypercholesterolemia

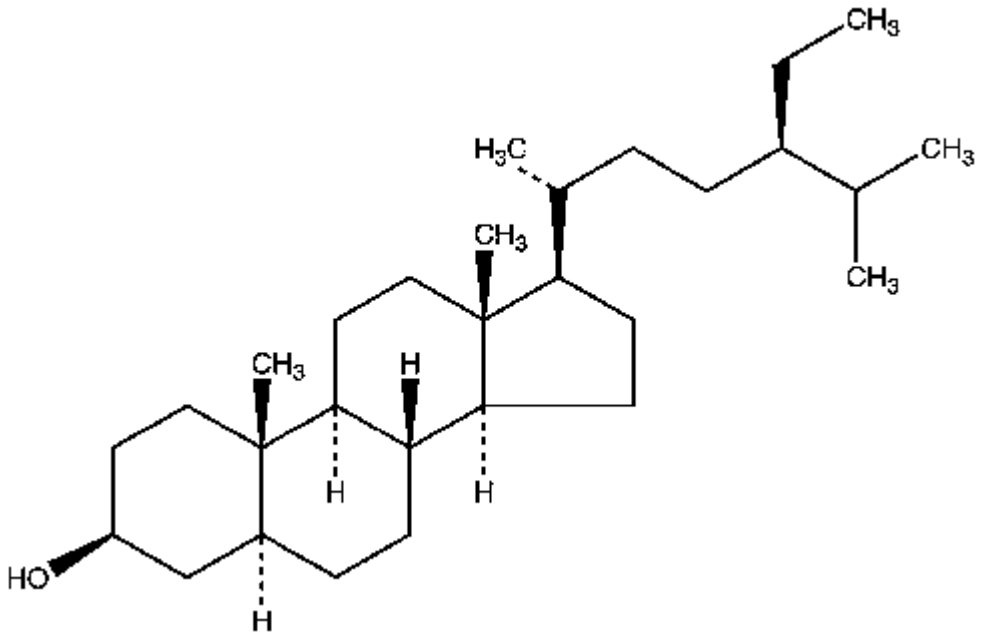
- Reduce intake of dietary cholesterol to less than 200mg/day
  - Proposed mechanism:
    - Reducing exogenous source of cholesterol reduces intracellular cholesterol pool and up-regulates LDL-receptors
  - Side effects: none

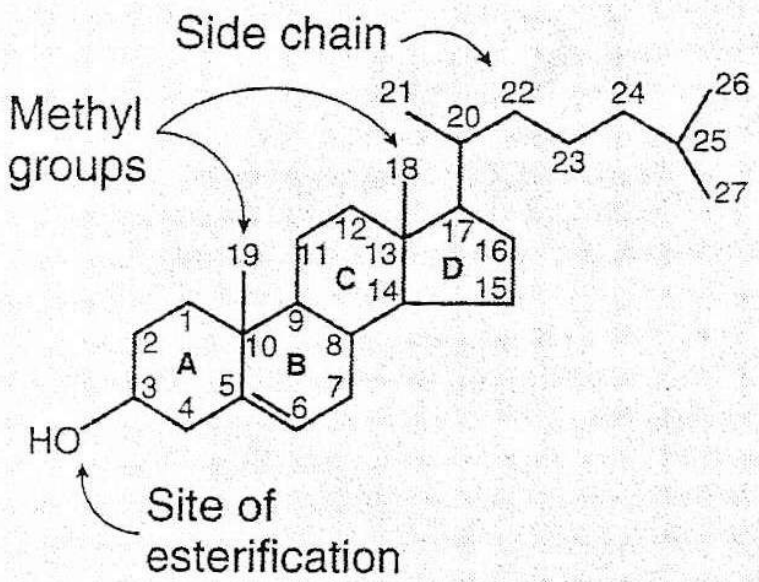
Nutritional and Pharmaceutical Means for Treating Hypercholesterolemia

- Increase consumption of viscous soluble dietary fiber (10-25g/d)
  - Proposed mechanisms:
    - Impairs absorption of dietary cholesterol
    - Impairs reabsorption of bile acids
    - Bacterial fermentation of soluble fibers results in short chain fatty acids that may inhibit cholesterol synthesis
  - Side effects: minimal (laxative)

Nutritional and Pharmaceutical Means for Treating Hypercholesterolemia

- Consume therapeutic doses of plant sterols and stanols (2g/d)
  - Functional foods
    - Benecol, Take Control
  - Proposed mechanism
    - Inhibit absorption of dietary cholesterol
    - Inhibit re-absorption of cholesterol in bile
  - Side effects: none

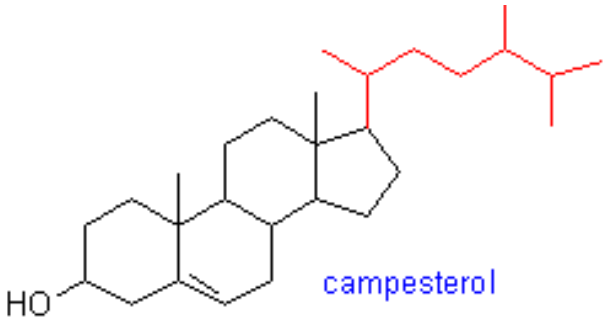


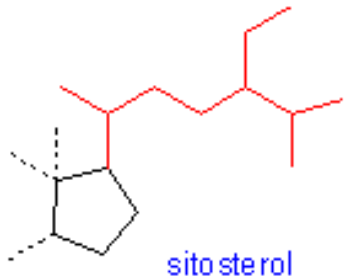


Cholesterol

Plant stanols

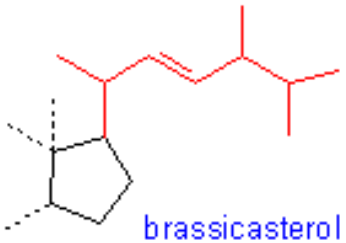
No double bond on B ring

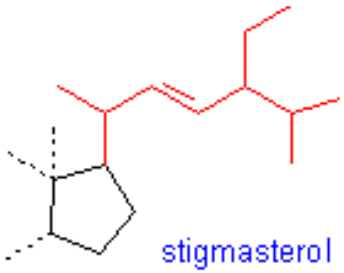


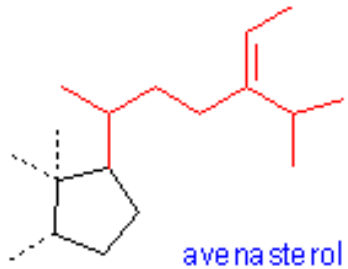


Plant sterols

Different side chains

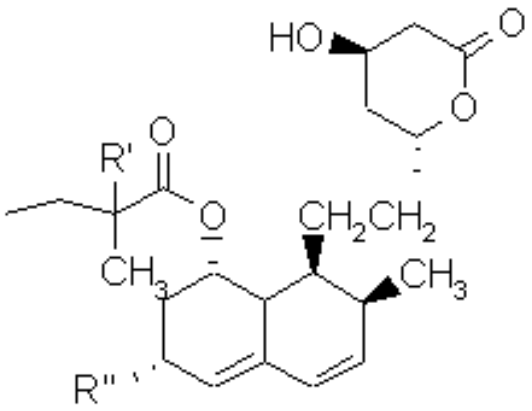




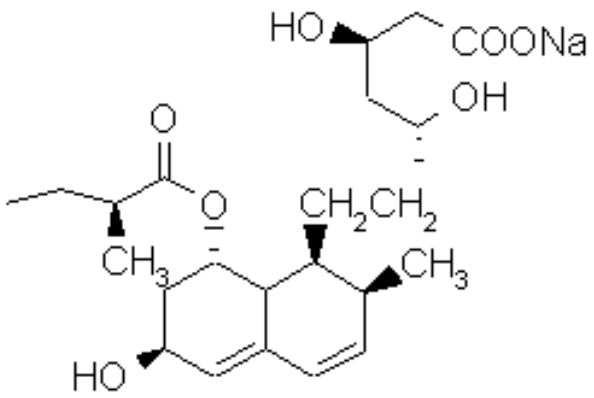


Nutritional and Pharmaceutical Means for Treating Hypercholesterolemia

- HMG-CoA Reductase Inhibitors
  - Statins
    - 18-55% reduction in LDL-C
    - Increases in HDL and decreases in TG
  - Proposed mechanism of action
    - Inhibition of cholesterol synthesis reduces intracellular cholesterol pool and up-regulates LDL-receptors
  - Side effects: myopathy, increased serum hepatic enzymes

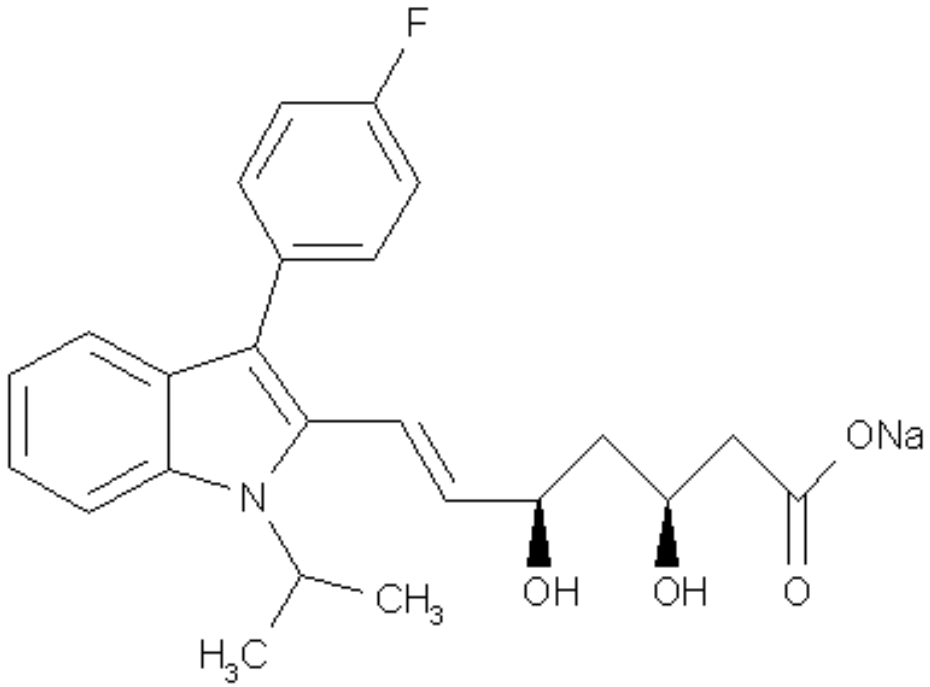


	R'	R''
Mevastatin	H	H
Lovastatin	H	CH_3
Simvastatin	CH_3	CH_3



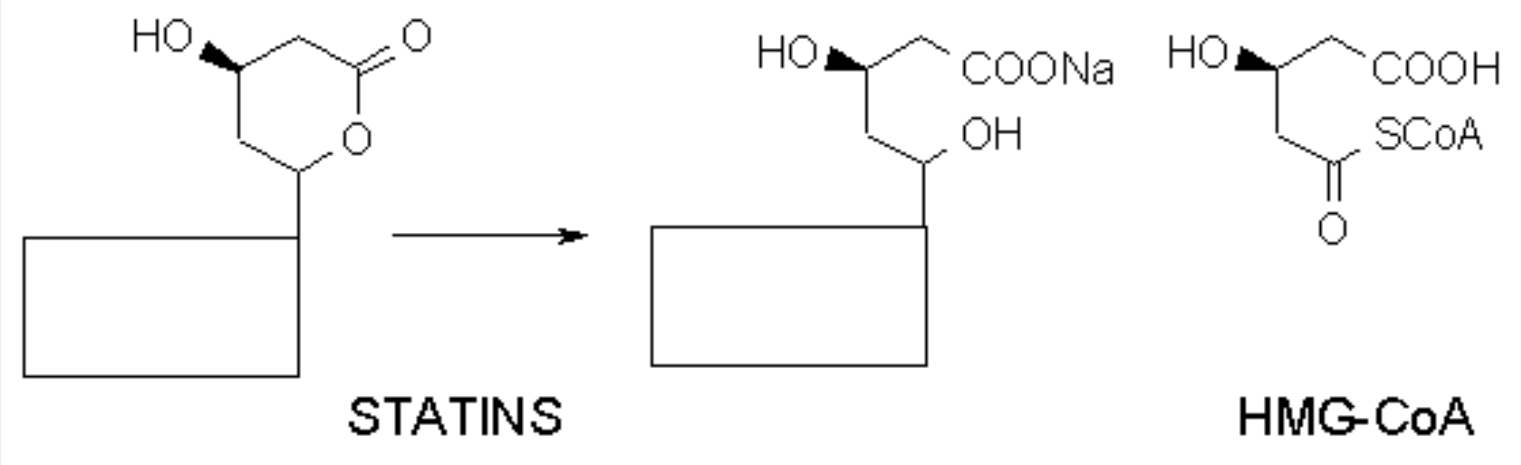
Pravastatin

Structures of Common statin drugs



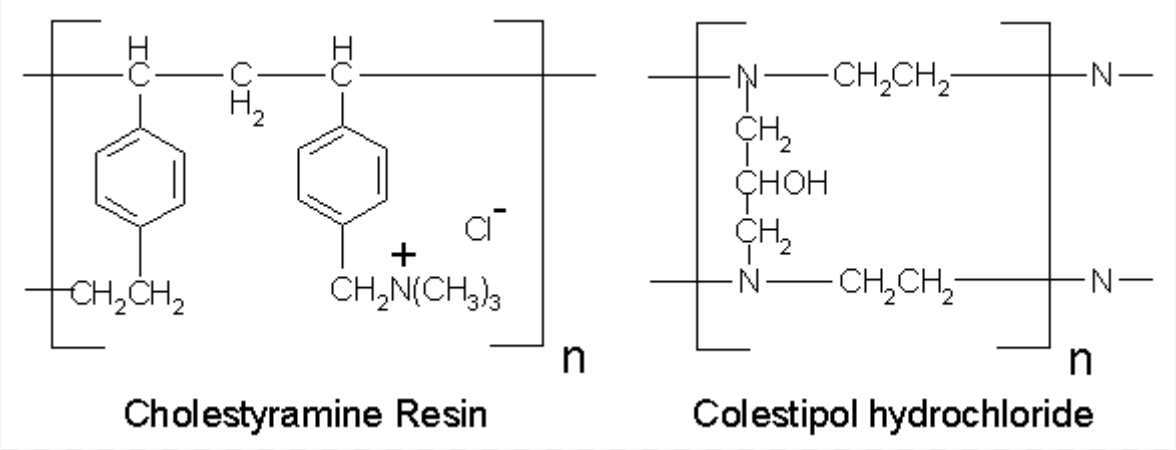
Fluvastatin sodium

Statin drugs are structural analogs of HMG-CoA



Nutritional and Pharmaceutical Means for Treating Hypercholesterolemia

- Bile acid sequestrants
  - Reduces LDL by 15-30%
  - Mechanism of action
    - Binds and prevents reabsorption of bile acids
    - Increases hepatic synthesis of bile acids, reduces cholesterol pool, up-regulates LDL-receptors
  - Side effects: GI distress, constipation, decreased absorption of other drugs



Cholestyramine Resin

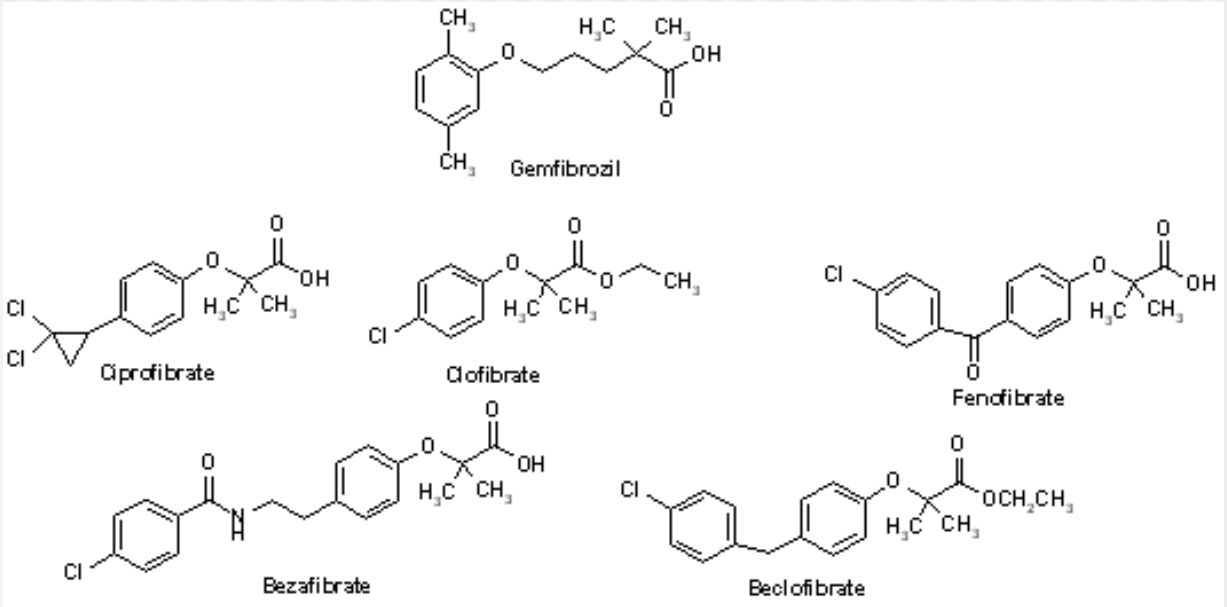
Colestipol hydrochloride

Nutritional and Pharmaceutical Means for Treating Hypercholesterolemia

- Pharmacological doses of niacin
  - 5-25% reduction in LDL
    - Increases HDL, decreases LDL
  - Proposed mechanism
    - Reduces VLDL synthesis
    - Decreases lipolysis in adipose
    - Increases LPL activity
    - Decreases esterification of TG in liver
  - Side effects: flushing, GI distress, hyperglycemia, hyperuricemia, hepatotoxicity

Nutritional and Pharmaceutical Means for Treating Hypercholesterolemia

- Fibric Acids
  - Decreases LDL by 5-20%
    - Larger decreases in TG (20-50%), increases HDL
  - Mechanism of action: increases LPL activity
  - Side effects: dyspepsia, myopathy, gallstones



Case Study
familial hypercholesterolemia

- 8 yo girl
  - Admitted for heart/liver transplant
- History
  - CHD in family
  - 2 yo xanthomas appear on legs
  - 4 yo xanthomas appear on elbows
  - 7 yo admitted w/ MI symptoms
    - [TC] = 1240 mg/dl
    - [TG] = 350 mg/dl
    - [TC]father = 355 mg/dl
    - [TC]mother = 310 mg/dl
  - 2 wks after MI had coronary bypass surgery
  - Past year severe angina & second bypass
  - Despite low-fat diet, cholestyramine, & lovastatin, [TC] = 1000 mg/dl

Xanthomas

- Raised, waxy appearing, often yellow skin lesions (shown here on knee)
  - Associated with hyperlipidemia
- Tendon xanthomas common on Achilles and hand extensor tendons







Xanthomas
raised lesions related to hyperlipidemia



Eruptive Xanthomas
-generally associated with hypertriglyceridemia



Xanthomas of the eyelid
-generally associated with hypercholesterolemia

Familial Hypercholesterolemia

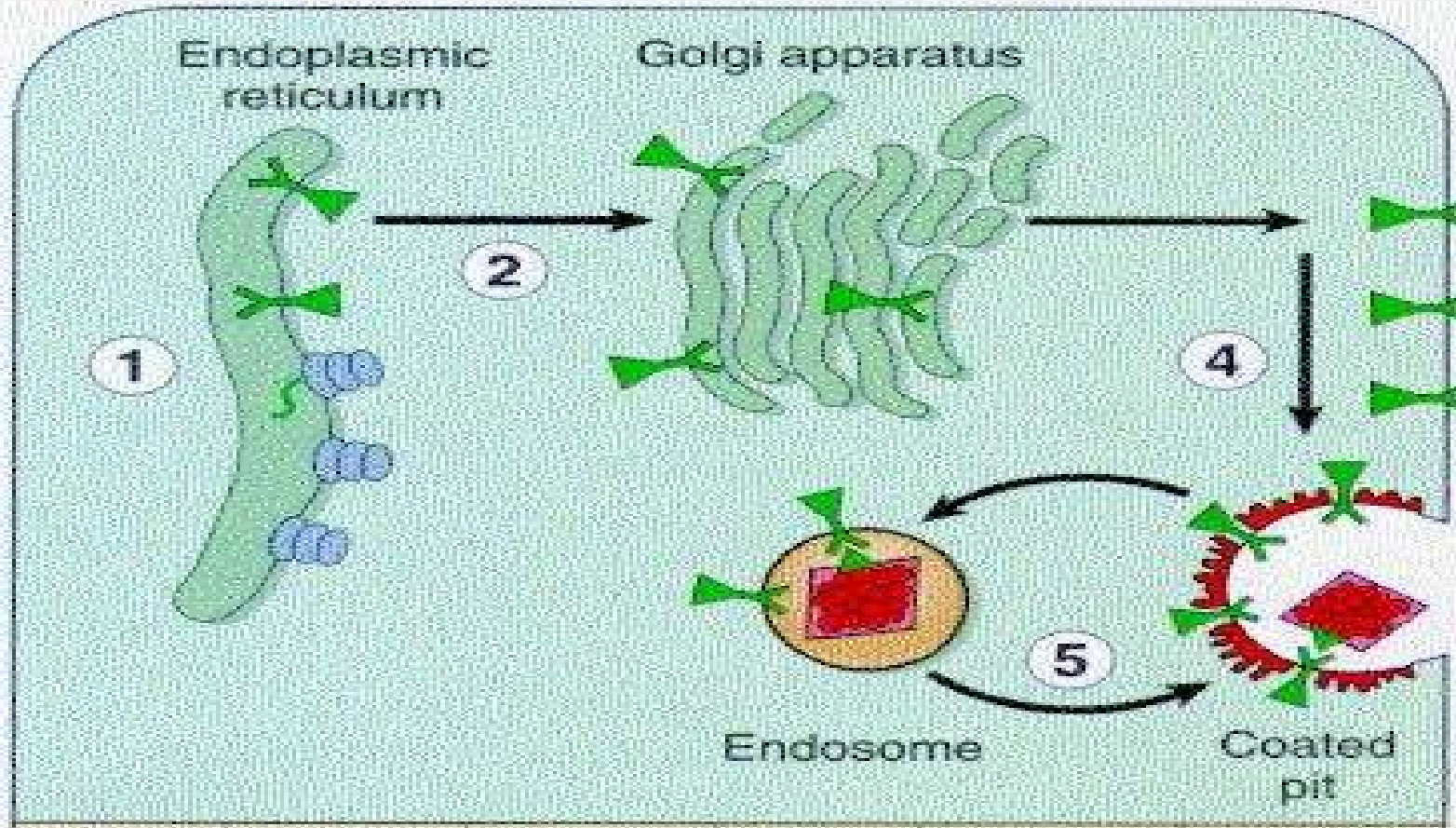
- LDL receptor deficiency
  - Gene for LDL-receptor on chromosome 19
    - No gender difference
    - Mutation is recessive
- Heterozygous FH
  - 1/500
    - French Canadians (1/270), Christian Lebanese (1/170), South African Afrikaners (1/100), Ashkenazi Jews (1/67, Jews descended from families from eastern Europe, comprise 80% of all Jews, higher risk for several diseases including breast, ovarian, colon cancers).
  - Hypercholesterolemia and premature CAD

Familial Hypercholesterolemia

- Homozygous FH
  - 1/1,000,000
  - Extremely high LDL-cholesterol
    - Xanthomas common
  - Very early symptomatic CHD

Familial Hypercholesterolemia

- LDL-receptor deficiency
  - 420 different mutations identified (dominant trait)
  - LDL-receptor activity: 0-25% of normal
- Classes of LDL-receptor mutations
  - 1 – no receptors
  - 2 – blockage of receptor from ER to Golgi Apparatus
  - 3 – receptor does not bind LDL normally
  - 4 – receptor does not accumulate in cathrin-coated pit
  - 5 – receptor fails to release LDL after internalization and does not recycle to cell surface



Mutation class	Synthesis	Transport	Binding	Clustering	Recycling
I	x				
II	→	x			
III	→	→	x		
IV	→	→	→	x	
V	→	→	→	→	x

Treatment of FH

- **Heterozygous FH**
- Dietary interventions, weight loss, exercise
  - Alone only moderately successful
- Cholesterol lowering-drugs
  - In combination with diet will cause up-regulation of LDL-receptors
  - Most powerful statins at highest dosage will result in ~60% reduction in LDL-C

Medications

- Bile Acid Sequestrants (Resins)
  - Anion exchange resins
    - Prevents reabsorption of bile salts
    - Effects additive when used with statins
    - May inhibit absorption of fat soluble vitamins (use multi-vitamin supplement)
  - Cholestryramine (Questran)
  - Cholestipol (Cholestid)
  - Cholesavelem (Welchol)
    - Newest resin, better tolerated than traditional resins

Medications

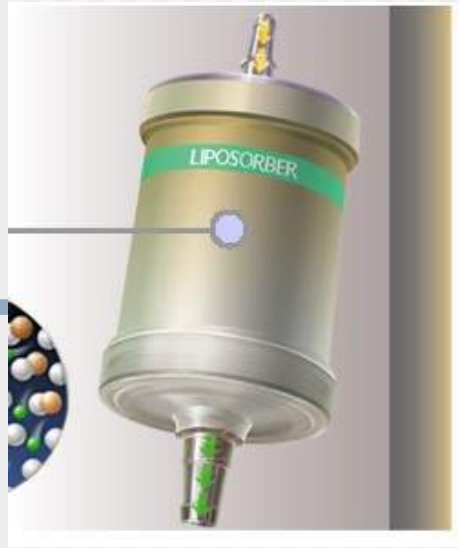
- HMG-CoA Reductase Inhibitors (statins)
  - Most potent LDL-C lowering drug
  - Modest TG lowering and HDL-C increasing effects
  - Atorvastatin (Lipitor)
  - Simvastatin (Zochor)

Treatment of FH

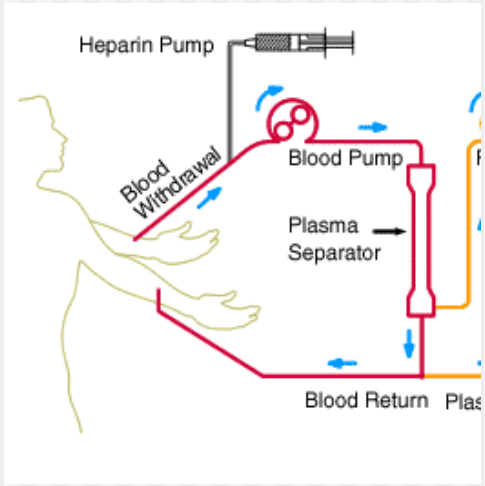
- **Homozygous FH**
- Diet, exercise, weight loss, drugs
  - Small to no effect on LDL-C
    - Dependent on activity of LDL-receptor

Treatment of homozygous FH

- LDL-apheresis
- Selective binding of apo B lipoproteins
  - LDL, VLDL, IDL, LP(a)
  - Dextran sulfate cellulose beads
- Reduces LDL-C by ~80%
- Used every 2 weeks
- FDA approved 1997
- $3000/treatment
  - Insurance coverage?







Treatment of homozygous FH

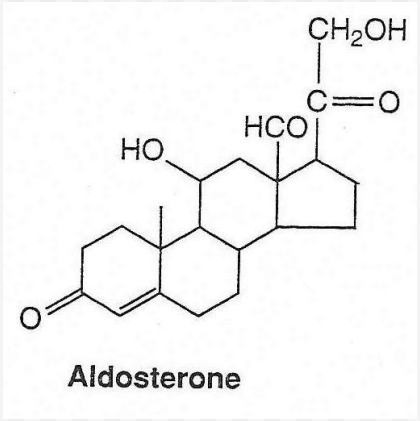
- Liver transplantation
  - ~70% of LDL-receptors in liver
  - High risk, long-term immuno-suppression, high cost
    - Success rate (Columbia Univ) 1yr – 92%, 5 yr – 88%

Resolution of Clinical Case

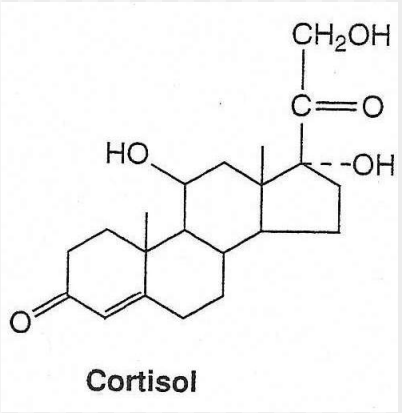
- Patient is homozygous for FH
  - Parents appear to be heterozygous FH
  - Early appearance of xanthomas
  - Cholestyramine and lovastatin treatment ineffective
  - Early symptomatic CHD
- Combined liver and heart transplant
  - Liver has ~ 70% of total LDL-receptors
  - Heart in FH often with significant CAD
- Transplantation was successful !!!
  - Good hepatic and cardiac function
  - TC = 26 mg/dl, regression of xanthomas

Steroid Hormone Metabolism: Adrenal Steroid Hormones

- Aldosterone
  - C21 derivative of cholesterol
  - Promotes renal
    - Sodium retention
    - Potassium excretion
- Glucocorticoids (cortisol)
  - Starvation
    - Hepatic gluconeogenesis
    - Muscle protein degradation
    - Adipose lipolysis
- Adrenal androgens
  - Dehydroepiandroterone (DHEA)
    - Precurser to potent androgens in extra-adrenal tissues



Aldosterone



Cortisol